

Multiple endocrine neoplasia type 2A

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Repurposing hypothesis: LENVATINIB → Multiple endocrine neoplasia type 2A

Repurposing-only pool: the candidate compounds for this target were restricted to FDA-approved / known drugs (ChEMBL max_phase $\geq$ 4) at collection time. Unapproved research-grade tool compounds were excluded from the pool, not merely down-ranked.

1. Hypothesis summary

LENVATINIB is proposed as a repurposing candidate against **Multiple endocrine neoplasia type 2A** via the target **RET**. It shows a ChEMBL median pChEMBL affinity of 8.82 at assay confidence 9/9, an Open Targets target-disease association of 0.860, and a Tanimoto similarity of 0.429 to CABOZANTINIB. Target network context (BioGRID, physical/genetic — not mechanism): STAT3, DOK6, DOK5, SRC, SHC1, PTPRF, DOK2, DOK4. The resulting composite score is 0.8357.

Chemist rationale: Lenvatinib has a median pChEMBL affinity of 8.82 against RET at an assay confidence score of 9 out of 9, and it is an approved drug with a Tanimoto similarity of 0.429 to the nearest approved drug cabozantinib. BioGRID records physical and genetic

interactors of RET as STAT3, DOK6, DOK5, SRC, SHC1, PTPRF, DOK2, and DOK4.

STAGE 1 PRIORITIZATION SCORES

- **tractability_score:** 0.5812 (ChEMBL bioactivity + AlphaFold pLDDT – prior-trial-failure penalty)
- **unmet_need_score:** 1.0000 (treatment availability + prevalence)

These are computed by the same formulas used to rank the full rare-disease / NTD universe; a manually chosen target is scored identically, never faked or skipped.

2. Evidence table

Evidence	Value
ChEMBL median pChEMBL affinity	8.82
Assay confidence score (0-9)	9
Open Targets association score	0.860
Tanimoto to nearest approved drug	0.429 (CABOZANTINIB)
Approved / known drug	yes
Mutation-specific approved indication (disclosure)	No specific mutation named in approved indication
Lipinski/Veber (MW, logP, HBD, HBA, TPSA, rotB)	426.9, 4.07, 3, 5, 115.6, 6
Lipinski violations / Veber pass	0 / yes
PubChem XLogP (lipophilicity)	n/a

Evidence	Value
AFDB apo structure mean pLDDT (free protein, no ligand)	78.8
Boltz structure confidence (0-1)	0.804
Boltz binding-pose confidence (0-1)	0.773
Boltz predicted affinity (relative optimization score, 0-1, NOT a Kd)	0.527
Boltz predicted structure (CIF)	Download CIF (/api/structures/e2c44d8a5ca3591638cd5eac3acbbc19.cif)
Boltz ADME – lipophilicity (logD)	2.612
Boltz ADME – permeability	0.538
Boltz ADME – solubility	medium-confidence
openFDA adverse-event signal (FAERS)	HYPERTENSION (1732), DIARRHOEA (1581), MALIGNANT NEOPLASM PROGRESSION (1574), FATIGUE (1242), DECREASED APPETITE (944)
Prior trials for this exact drug+disease	0
Target discovery method	genetic_association

3. Full source citations

- **PMIDs (7):** 16343738, 24699901, 28931560, 29175871, 29348307, 33812987, 39067973
- **ChEMBL activity IDs (1):** 15247904
- **NCT numbers (0):** none

4. Composite score breakdown

Term	Weight	Component value	Contribution
Normalized pChEMBL affinity	0.30	0.831	0.2494
Assay confidence (score / 9)	0.20	1.000	0.2000
Normalized Open Targets association	0.20	0.860	0.1720
Normalized Tanimoto similarity	0.15	0.429	0.0643
No prior failed trial (1/0)	0.15	1	0.1500
Composite (weighted sum - penalty - cap)			0.8357

Weighted sum before penalty = 0.8357; penalty = 0.0000; reported composite_score = 0.8357.

5. Limitations

- **Binding is not efficacy.** A high binding-pose confidence (0.773) or predicted affinity (0.527) only suggests the molecule may occupy the target; it does NOT establish agonism vs. antagonism, functional modulation, or therapeutic benefit.
- **ADME values are model predictions, not measurements.** The Boltz lipophilicity/permeability/solubility numbers are computed estimates and must be confirmed experimentally before any decision.
- **Structure confidence is bounded.** This hypothesis relies on a Boltz structure_confidence of 0.804 (complex pLDDT 0.759) and an AFDB apo mean pLDDT of 78.8; the AFDB model contains NO ligand, so the protein-ligand pose is entirely a Boltz prediction.
- **Assay-type and species caveats.** The affinity is a median pChEMBL over Homo sapiens IC50/Ki assays at confidence ≥ 8 ; assay heterogeneity and the bounded approved-drug reference set for Tanimoto still apply.

- **Absence of evidence is not evidence of absence.** A zero prior-trial count or no adverse-event signal may reflect that the pair has simply never been studied, not that it is safe or untried.
- **This is a repurposing *hypothesis*, not a finding.** It is a prioritised starting point that requires wet-lab and, ultimately, clinical validation.

TARGET DRUGGABILITY CONTEXT

- **Approved drugs with known mechanism against this target (ChEMBL):** 9 — ALECTINIB HYDROCHLORIDE, PRALSETINIB, QUIZARTINIB, REGORAFENIB, SELPERCATINIB
- **Historical difficulty literature:** insufficient signal found (fewer than 2 qualifying abstracts in targeted PubMed searches for undruggability / resistance / difficulty).

Druggability context is informational only. It does not affect tractability_score, unmet_need_score, composite_score, or STRONG_MATCH.

HUMAN REVIEW CHECKPOINT

This is a falsifiable hypothesis, not a finding.

Record your scientific reasoning below, then sign off. Approving advances this candidate for wet-lab validation; rejecting closes the case permanently.

REVIEWER NOTE **required**

e.g. Affinity and structure confidence justify wet-lab follow-up despite the sub-threshold composite score.



Approve case

Reject case